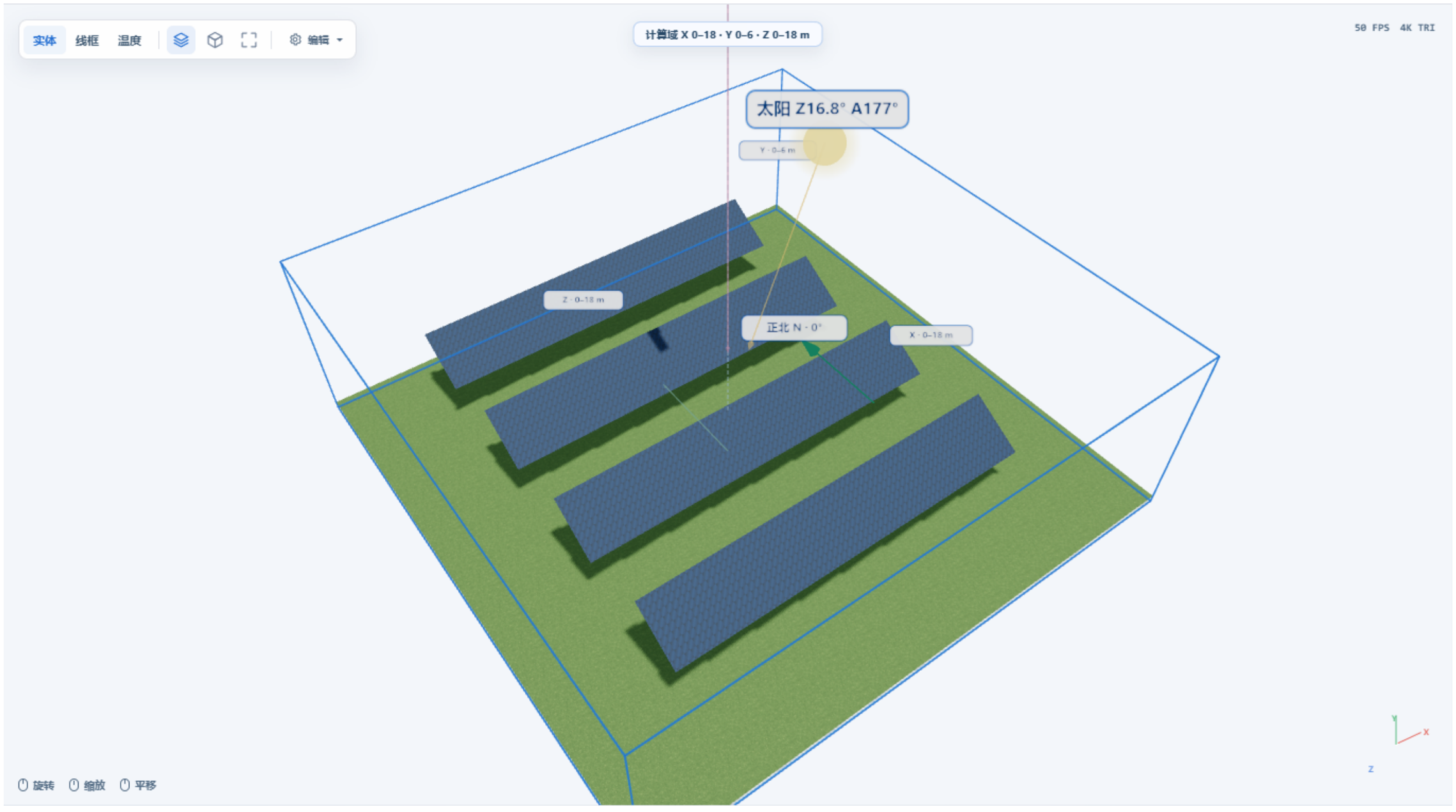
光伏板下的草地白天会更凉吗?

板面在升温，草叶却可能出现不同的响应。
用同一气象下的两片草场，观察热量如何分配。



12:00 · 基础网格模拟

46.2°C

光伏板平均物理温度

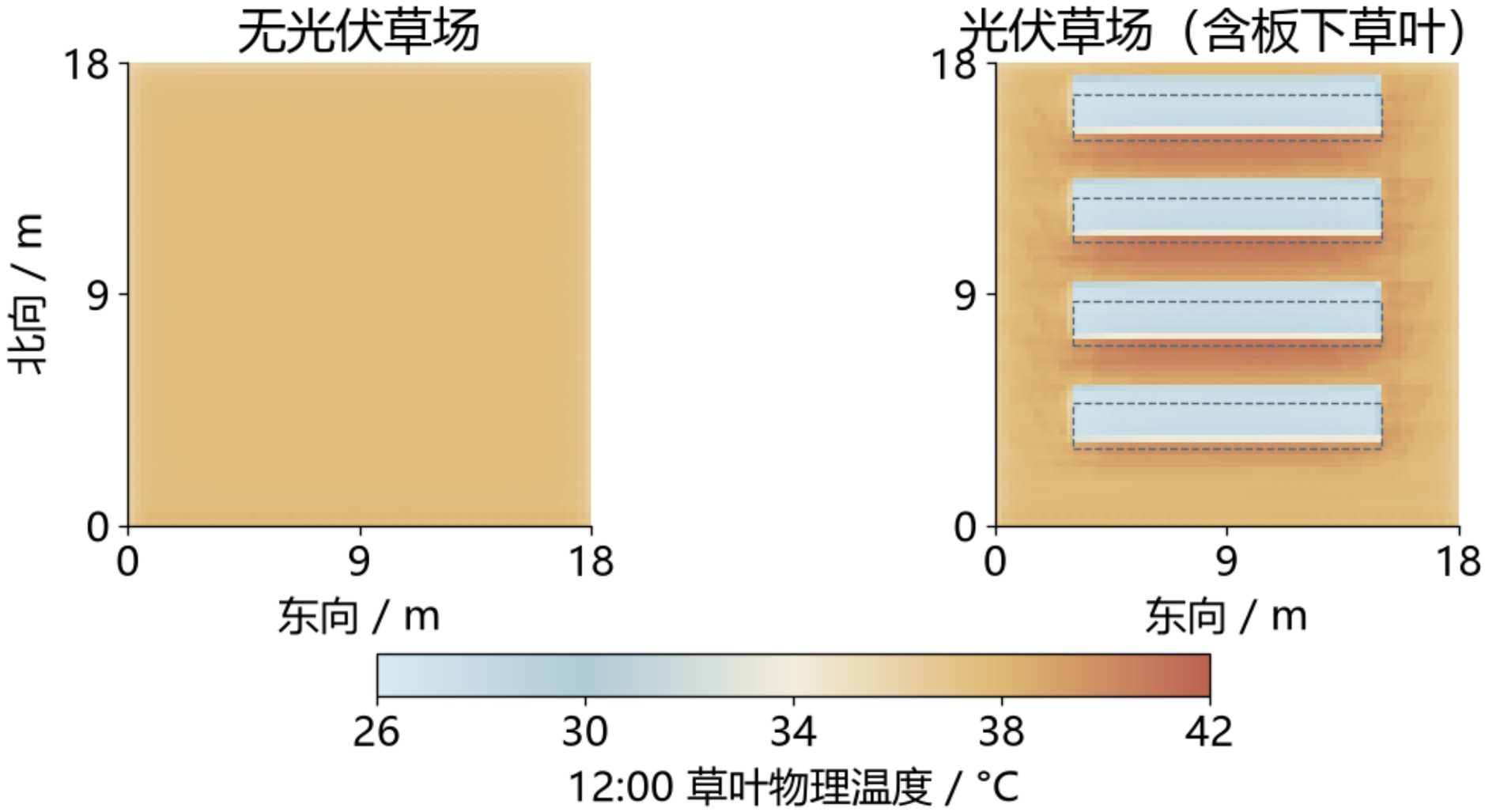
37.6 → 35.2°C

无光伏 → 有光伏草叶均温

18 × 18 m草场 · 4行南向30°阵列 · 96 m²组件
研究背景参考[1,2]；图中数值来自本案例原生模拟。

代码：github.com/bianzj/streamstudio　程序：github.com/bianzj/streamstudio/releases/tag/v2.0.1
百度网盘：pan.baidu.com/s/1iYGVT5aC67xQ9T5A00A_YA?pwd=uvx2　提取码：uvx2

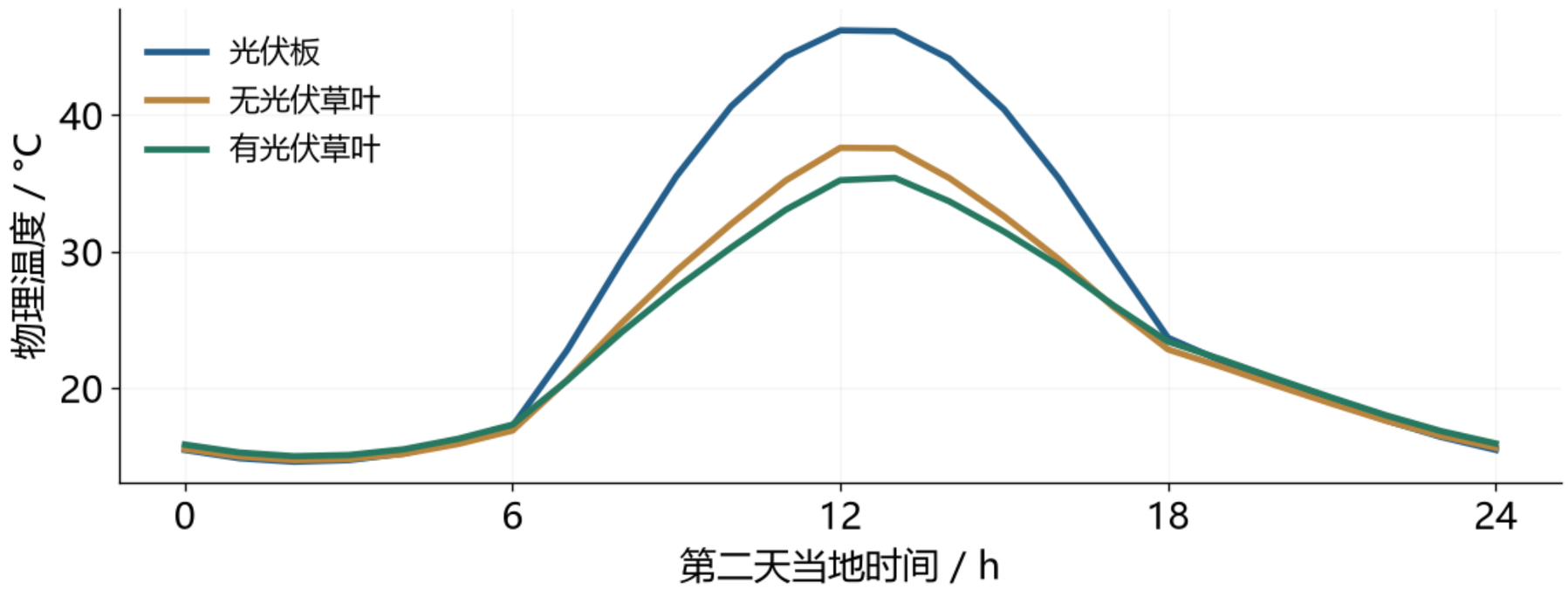
同一片草场，板下与行间怎样变化?



午间草叶均温变化 −2.32 K

0.25 m原生网格独立补算；虚线是板的垂直投影。

每格平均两片原生三角面元，没有空间插值。



上图是0.25 m午间空间诊断；全天曲线沿用1 m草叶网格。

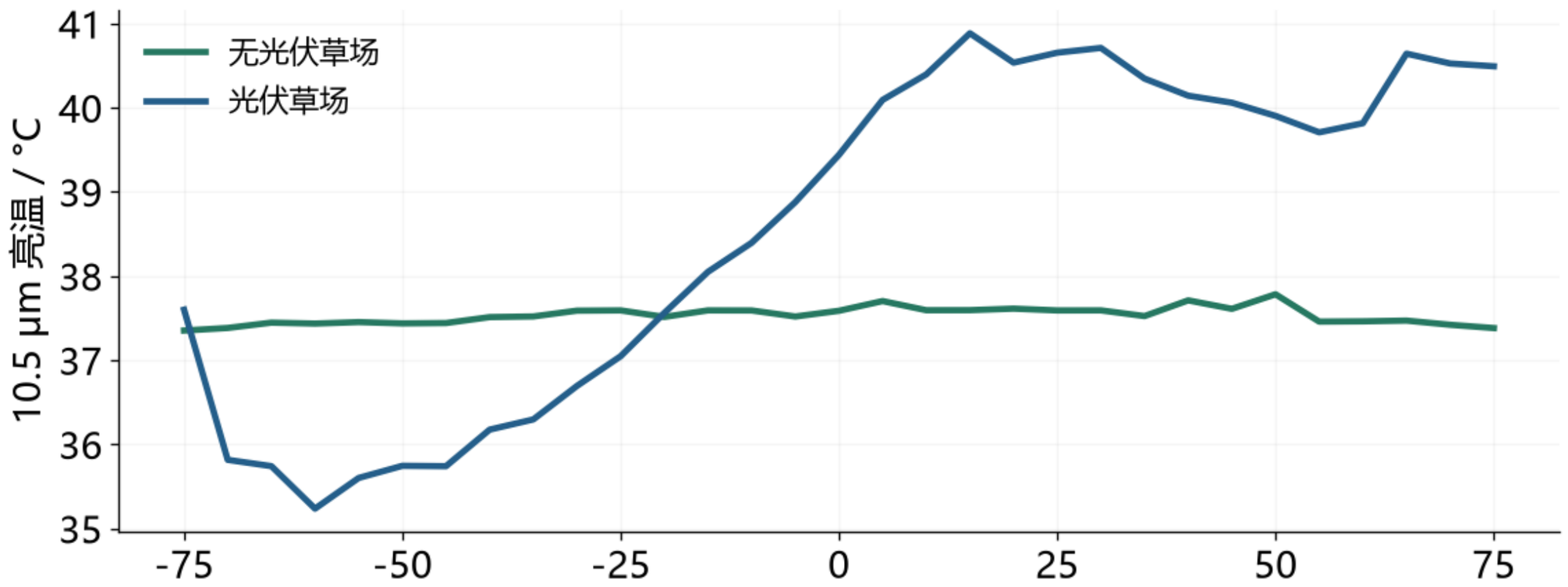
前24个节点预热。草叶温差不能直接解释为气温变化。[1,2,5]

代码：github.com/bianzj/streamstudio　程序：github.com/bianzj/streamstudio/releases/tag/v2.0.1

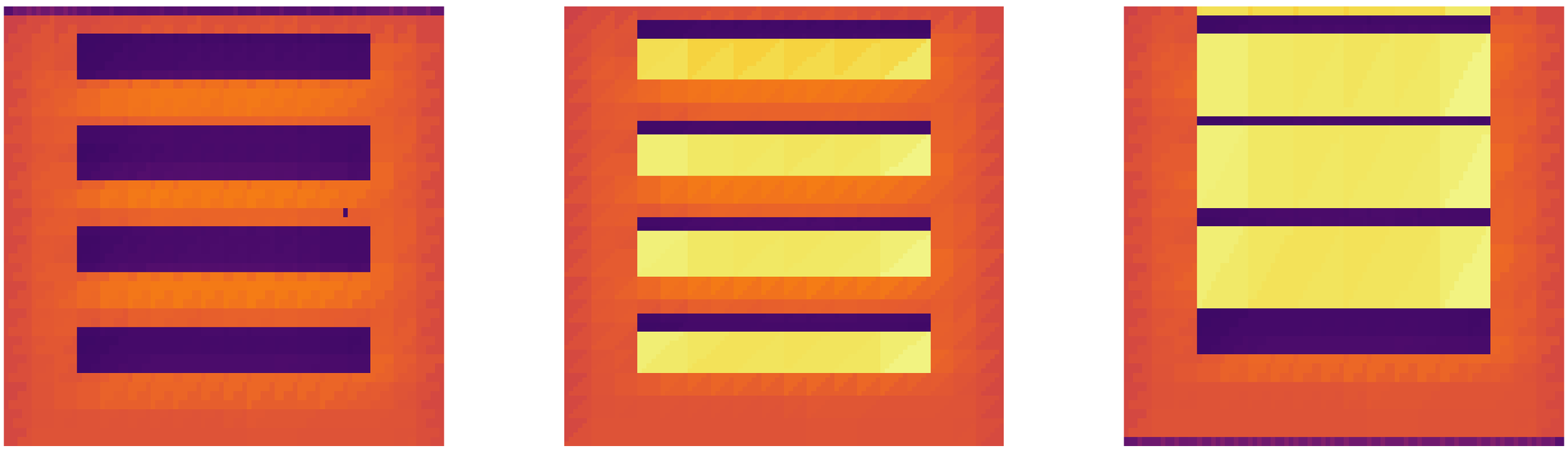
百度网盘：pan.baidu.com/s/1iYGVT5aC67xQ9T5A00A_YA?pwd=uvx2　提取码：uvx2

换一个角度看，热红外亮温也会变

同一温度场 · 12:00 · 10.5 μm
板面、草叶与土壤的可见比例随方向改变。[4]



61个方向，场景亮温极差约 5.65 K

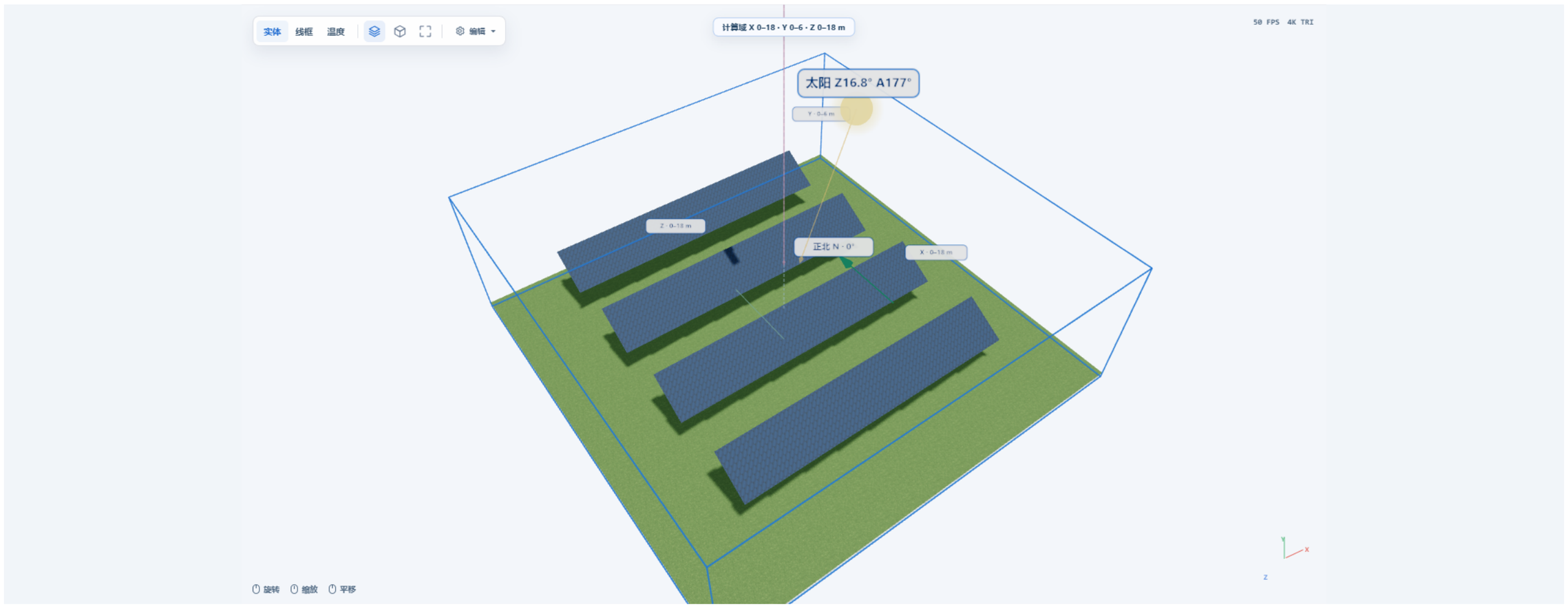


北向60°　　天底0°　　南向60°

三幅影像使用同一22–48°C色标。曲线示南北观测平面。
亮温不是板的物理温度；板发射率固定取0.90，未验证角度依赖发射率。

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百度网盘：pan.baidu.com/s/1iYGVT5aC67xQ9T5A00A_YA?pwd=uvx2　提取码：uvx2

从三维场景，到温度与能量的解释



面元能量平衡同时考虑辐射交换、显热、储热与发电耗能。
板温由原生模型求解，未用Faiman经验式替代。[3]

参考文献

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[2] Barron-Gafford, G. A., et al. (2019). Agrivoltaics provide mutual benefits across the food–energy–water nexus in drylands. Nature Sustainability, 2, 848–855. doi:10.1038/s41893-019-0364-5

[3] Faiman, D. (2008). Assessing the outdoor operating temperature of photovoltaic modules. Progress in Photovoltaics: Research and Applications, 16(4), 307–315. doi:10.1002/pip.813

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[5] Barron-Gafford, G. A., Minor, R. L., Allen, N. A., et al. (2016). The Photovoltaic Heat Island Effect: Larger solar power plants increase local temperatures. Scientific Reports, 6, 35070. doi:10.1038/srep35070

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